Lexical Tone and Word Order Matter in Translation: Prosodic Correspondence Across Languages

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Abstract

Prosody conveys emphasis, emotion, and intent, yet its correspondence across translation remains underexplored, especially when languages differ in lexical tone and word order. To our knowledge, we present the first systematic word-level analysis of prosodic correspondence in translated speech spanning tonal and non-tonal Asian languages, with comparisons to European-language findings and explicit analyses of lexical tone and word reordering. Building on prior evidence of fine-grained prosody correspondence in European-language pairs, we investigate how these linguistic factors relate to word-level pitch and energy correspondence between English and five languages: Mandarin Chinese, Thai, and Vietnamese, which have lexical tone, and Japanese and Korean as additional comparisons. Correspondence is generally lower than previously reported European-language estimates. Adjusting for lexical tone increases pitch correspondence for English–Mandarin Chinese, English–Thai, and English–Vietnamese. However, English–Japanese and English–Korean also exhibit lower correspondence than the European-language estimates, indicating that tonal status alone does not explain the observed pattern. Greater word reordering is associated with weaker local correspondence even under semantic alignment, suggesting that semantic alignment does not fully account for differences in prosodic organization. These findings characterize the limits of fine-grained acoustic correspondence across translation and motivate prosody modeling that accounts for lexical pitch constraints and the structural context of translated content.

1 Introduction

Prosody conveys emphasis, emotion, and discourse structure beyond lexical content (Gussenhoven, 2004; Cutler et al., 1997). Preserving this information is an important goal of expressive speech-to-speech translation (Seamless Communication et al., 2023; Swiatkowski et al., 2023; Ulgen et al., 2026). However, languages differ in how pitch encodes lexical distinctions and how translated content is ordered. Effective prosody transfer therefore requires understanding how these differences affect cross-lingual correspondence.

Professionally dubbed speech provides a useful setting for studying this correspondence. Dubbing adapts dialogue to another language while maintaining its role in the same scene and meeting expressive and temporal constraints (Brannon et al., 2023). Using dubbed speech, Xie et al. (2026) report positive word-aligned pitch and energy correlations for German–English, English–Spanish, and English–French. However, their study is limited to European-language pairs without lexical tone. It also reorders acoustic segments to align corresponding words, although lexical order is generally similar in the studied pairs. It therefore remains unclear whether these findings generalize to languages with lexical tone or substantial word reordering, and how much the measured correlations reflect local prosodic correspondence versus broader utterance-level patterns.

The first question concerns lexical tone. Pitch in tonal languages contributes to lexical distinctions as well as sentence-level prosody (Gussenhoven, 2004; Xu, 1999). The lexical function of pitch might compromise the correspondence; semantically corresponding words may therefore have different pitch contours even when they serve similar expressive functions. If lexical tone contributes to lower measured correspondence, reducing tone-related variation should increase the similarity of aligned contours. Comparisons beyond tonal languages are also necessary to assess whether lower correspondence is specific to lexical tone. We therefore examine English paired with Mandarin Chinese, Thai, and Vietnamese, alongside Japanese and Korean. Japanese introduces a lexical pitch-accent system, while Korean provides an additional

comparison beyond the three tonal languages.

The second question concerns how differences in sentence structure across languages relate to prosodic correspondence. Word-aligned analysis brings semantically corresponding units together, enabling local comparisons. However, language pairs with greater structural difference with different word-orders may exhibit greater reordering of corresponding regions, while language-specific segmentation may lead to lower alignment coverage. In such pairs, fewer regions may be aligned, and corresponding regions may occur in different positions within the utterance. Lower word-aligned correlation may therefore reflect both differences in prosodic realization and cross-language variation in word order and alignment coverage.

To our knowledge, this is the first systematic word-level study of prosodic correspondence across tonal and non-tonal Asian languages to examine both lexical tone and word reordering. We analyze pitch and energy for English paired with Mandarin Chinese, Thai, and Vietnamese, which have lexical tone, and with Japanese and Korean. We also compare all three tonal-language pairs directly. Episode-based cross-fitting estimates tone-related pitch variation, and matched-utterance comparisons measure the effect of adjustment. We assess alignment coverage and compare semantic and temporal pairing to examine how word reordering relates to measured correspondence, using a French reference processed with the same pipeline.

All examined pairs show positive mean pitch and energy correlations above unit-shuffled baselines. In contrast to the similar pitch correlations reported across three European-language pairs by Xie et al. (2026), we find lower pitch correspondence for all five English–Asian pairs. Energy correspondence above baseline is also lower. Tone adjustment increases pitch correspondence for English–Mandarin Chinese, English–Thai, and English–Vietnamese, with all gain intervals excluding zero, but adjusted means remain below both published European estimates and the current French reference. The gap therefore persists after tone adjustment and also occurs for Japanese and Korean, showing that the earlier findings do not generalize uniformly across languages and that tonal status alone does not explain the difference. Direct tonal-language comparisons show adjustment gains for Mandarin Chinese–Vietnamese and Vietnamese–Thai, but no clear change for Mandarin Chinese–Thai.

Japanese and Korean have the greatest mean reordering distances, lower alignment coverage, and the largest temporal-over-semantic pairing gains among reordered utterances. Greater reordering is associated with a larger temporal advantage for Vietnamese, Thai, Japanese, and Korean. Pitch correlation also tends to increase with alignment coverage, but coverage differences do not fully explain the lower correspondence: at 60–80% coverage, French remains at 0.223, compared with 0.094–0.155 for the Asian-language pairs, and reweighting to the French coverage distribution does not close the gap. Semantic-pairing correlations remain positive across the Asian-language pairs, but are lower than the French reference. Together, these results highlight that word reordering and alignment coverage are important context for interpreting prosodic correspondence across language pairs.

Our contributions are: (1) a systematic word-level analysis of prosodic correspondence across tonal and non-tonal Asian languages, including direct tonal-language comparisons and a European reference processed with the same pipeline; (2) cross-fitted lexical-tone adjustment showing that tone-related variation contributes to pitch mismatch but tonal status alone does not explain the observed differences; and (3) evidence that the degree of word reordering between different languages interacts with sentence-level pitch trends and semantic alignment to shape measured correspondence.

2 Related Work

2.1 Prosody in Speech Translation

Speech-to-speech translation has progressed from cascaded systems to direct models that jointly learn translation and speech generation (Jia et al., 2019, 2022; Inaguma et al., 2023). Alongside improvements in translation quality and naturalness, several studies have explicitly addressed the preservation of expressive information. Early work investigated the transfer of word-level emphasis (Do et al., 2017), while subsequent approaches explored cross-lingual prosody transfer for expressive dubbing and multilingual speech translation (Swiatkowski et al., 2023; Seamless Communication et al., 2023). These approaches motivate a closer examination of which prosodic characteristics can be preserved across languages.

A complementary line of research examines whether translation systems use prosody to in-

terpret source speech. Studies of Korean wh-phrases and contrastive benchmarks show that direct speech-to-text translation models can exploit prosody, although their sensitivity remains limited (Zhou et al., 2024; Tsiamas et al., 2024). These studies concern the contribution of source prosody to translated meaning. Our work instead examines the acoustic correspondence between source speech and human translations, providing empirical evidence relevant to the prosodic patterns that expressive S2ST systems aim to preserve.

2.2 Cross-Lingual Prosodic Correspondence

Human speech provides a complementary perspective on cross-lingual prosody. Brannon et al. (2023) analyze professional dubbing and find evidence that source audio influences translated speech beyond its lexical content. Using bilingual conversational reenactments, Avila and Ward (2023) investigate cross-language prosodic differences and develop a feature-based measure of prosodic dissimilarity. These studies establish useful settings for examining how expressive characteristics are maintained or adapted across languages.

A recent study measures fine-grained pitch and energy correspondence in professionally dubbed English–German, English–Spanish, and English–French speech (Xie et al., 2026). By comparing contours over aligned linguistic units, it finds cross-lingual correspondence above shuffled baselines. However, its analysis is restricted to non-tonal European languages with similar structure. We investigate prosodic correspondence involving languages with significant structural differences, such as Mandarin Chinese, Thai, and Vietnamese, where lexical tone adds a lexical function to pitch and sentence structures and word order may differ substantially. We also examine sensitivity to broad utterance-level pitch trends and alignment choices, distinguishing these factors from the local correspondence measured by the analysis.

2.3 Lexical Tone and Intonation

Lexical tone and intonation jointly shape pitch in tonal languages (Gussenhoven, 2004): pitch distinguishes lexical meaning while also conveying prominence, phrasing, and communicative function. Studies document these interactions in Mandarin, where tone and focus jointly shape F0 contours (Xu, 1999); Thai, where tonal realization varies by phrasal position (Morén and Zsiga, 2006); and Vietnamese, where lexical tones interact with intonational patterns associated with communicative functions (Ha and Grice, 2010). Thus, tonal contours in connected speech reflect prosodic context as well as lexical patterns.

This motivates examining local pitch correspondence in translations involving tonal languages without assuming that lexical tone necessarily reduces correspondence. Whereas prior studies characterize tone–intonation interactions within individual languages, we examine how English source contours correspond to translated speech in three tonal targets. Comparing pitch and energy further tests whether cross-lingual differences are specific to pitch or extend to another acoustic dimension.

3 Data Processing and Analysis Pipeline

We adapt the alignment-based analysis of Xie et al. (2026) to Chinese, Vietnamese, Thai, Japanese, and Korean. The pipeline combines speech filtering, timed transcription, bilingual alignment, and prosodic contour comparison. For Chinese, Vietnamese, and Thai, we additionally estimate and remove tone-related pitch variation. Japanese and Korean provide further comparisons for assessing whether lower correspondence extends beyond these tonal languages.

3.1 Data Extraction and Filtering

Speech Data. We analyze professionally dubbed television dialogue mined using SpeechVecAlign (Meng and Koehn, 2025). Each pair contains semantically corresponding utterances extracted at their respective track timestamps and converted to 16 kHz mono. English serves as the common comparison language, although the English recording may itself be a dub.

We use all available Chinese, Vietnamese, and Korean pairs, and sample 100,000 Thai and 60,000 Japanese pairs. We also process 10,000 English–French pairs with the current settings. The 2,713 retained pitch-scored pairs provide a reference alongside the published European-language results.

Audio and Semantic Filtering. Prosodic analysis can be sensitive to several factors, including audio quality, overlapping speech, multiple speakers within an utterance, and inconsistencies in semantic alignment. We therefore filter the dubbed data along these dimensions. We apply SQUIM (Kumar et al., 2023) to both recordings and retain pairs meeting the thresholds in Table 1. We use

pyannote diarization (Bredin et al., 2020) to exclude pairs with more than one detected speaker in either recording. After transcription, we compute SONAR cosine similarity between the transcripts (Duquenne et al., 2023) and retain pairs scoring at least 0.45. We repeat the analysis at 0.63 as a semantic-filtering sensitivity test. Tables 2 and 3 summarize the retained data.

Stage	Metric	Threshold
Audio quality	SI-SDR	≥ 7.5
Audio quality	STOI	≥ 0.91
Audio quality	PESQ	≥ 1.25
Diarization	Speakers	$= 1$
Semantic filtering	SONAR cosine	≥ 0.45

Table 1: Filtering thresholds. Audio-quality and diarization criteria apply to both recordings.

3.2 Transcription and Word Alignment

Timed Transcription. We transcribe each recording using `whisper_timestamped` (Louradour, 2023; Radford et al., 2023), specifying the language of each recording.

Language-Specific Segmentation. We convert Chinese text to simplified characters using OpenCC (OpenCC contributors, 2026) and segment with spaCy's `zh_core_web_sm` (Honnibal et al., 2020). Japanese uses `ja_core_news_sm`, and Thai uses PyThaiNLP's `newmm` tokenizer (Phatthiyaphaibun et al., 2023). Vietnamese retains space-separated syllabic units. For Korean, Kiwi morphemes (Lee, 2024) are used for alignment and mapped back to the space-separated timed units for acoustic extraction.

Timestamps are retained when segmentation leaves a timed unit unchanged. Only units requiring subdivision receive estimated internal boundaries: their duration is distributed across characters, and each resulting segment receives the span of its constituent characters. The original timed unit determines the outer boundaries of these segments.

Bilingual Word Alignment. We run FastAlign (Dyer et al., 2013) in both directions with a diagonal prior and retain intersecting links. Training is augmented with TED parallel text from OPUS (Tiedemann, 2012): TED2013 for English–Chinese and English–French, and TED2020 for the remaining English pairs. Each background is capped at 143k retained sentence pairs and tokenized consistently with the speech transcripts.

3.3 Prosodic Contour Construction

Pitch and Energy. We extract pitch using YAAPT through pYAAPT (Kasi and Zahorian, 2002). Unvoiced regions are interpolated to obtain continuous contours. Energy is computed as RMS amplitude computed with librosa (McFee et al., 2015), using a 25 ms window and 10 ms hop. Values below the fifth percentile are set to zero.

Alignment Clusters. We group transcript units connected by bilingual alignment links into clusters, rather than grouping words by adjacency. A cluster may contain one or several units on either side. Using their timestamps, we extract each side's acoustic frames and resample them to 20 frames. We then concatenate paired chunks in the order of each cluster's first English-side unit, placing corresponding content at matching positions despite differences in word order.

We normalize positive feature values by their utterance-level mean to reduce speaker-dependent effects while preserving prosodic structure, since prosodic prominence is often conveyed relative to surrounding words and phrases. Pairs with constant or otherwise invalid score sequences are excluded. Contour-resolution and track-quality checks are reported in Appendix E.3.

3.4 Correlation Analysis

Correlation Metric and Baseline. For each utterance, we compute zero-lag Spearman rank correlation (Spearman, 1904) between the paired contour sequences and report the mean across finite scores. Prosodic patterns are meaningful in context, so we focus measuring the correspondence of the relative ordering of values rather than their absolute magnitudes. Pitch and energy means use their respective valid cohorts. Paired comparisons retain utterances with valid scores in both conditions.

The shuffled baseline randomly permutes target clusters once within each utterance before scoring. We report its mean alongside the observed correlation and calculate correspondence above baseline from paired utterance-level differences.

Statistical Analysis. We estimate 95% confidence intervals using 2,000 episode-clustered bootstrap draws. For paired contrasts, resampling retains the pairing between conditions. Reported changes are calculated from utterance-level differences rather than subtraction of corpus means.

Stage	EN–FR	EN–ZH	EN–VI	EN–TH	EN–JA	EN–KO
Selected pairs	10,000	6,720	66,352	100,000	60,000	47,308
Extracted pairs	9,994	6,720	66,352	99,932	60,000	47,308
After SQUIM	4,029	3,314	35,050	50,047	26,811	17,643
After audio gates	2,923	2,641	26,332	38,342	19,775	14,127
After SONAR	2,713	1,842	22,352	31,281	15,321	11,452
After pitch scoring	2,713	1,838	22,332	31,261	15,267	11,350

Table 2: Utterance pairs retained at each processing stage. Audio gates include diarization. Pitch scoring excludes pairs without valid, nonconstant contour sequences.

Pair	Utterances	Episodes	Hours
EN–FR	2,713	1,324	5.4
EN–ZH	1,838	55	2.6
EN–VI	22,332	446	37.8
EN–TH	31,261	1,132	54.5
EN–JA	15,267	1,453	24.6
EN–KO	11,350	411	17.2

Table 3: Final pitch-scored cohorts. Approximate audio hours are summed across both recordings. French denotes the reference processed with the current settings.

3.5 Lexical Tone Adjustment

We assess the contribution of lexical tone by estimating tone-related pitch variation and repeating the contour comparison after normalizing and adjusting according to the estimated lexical tone.

Tone Labels and Timing. We derive syllable-level tone labels using pypinyin for Mandarin (pypinyin contributors, 2025), orthographic rules and pronunciation exceptions for Thai, and written tone markings for Vietnamese. MMS forced alignment (Hermjakob et al., 2018; Pratap et al., 2023) locates romanized syllables in the audio. Successive syllable onsets define the intervals used by the tone model.

Tone Model. We estimate typical pitch patterns from each syllable's tone label and those of adjacent syllables not separated by a pause, using ridge-regularized linear regression. Pitch is expressed in semitones relative to the utterance's median F0. The model represents each tone's contribution as a smooth curve over syllable time that varies with syllable duration. This captures both a high tone's pitch offset and a falling tone's changing contour. The model also includes tone-pair interactions, utterance position, and pauses. We subtract only the centered tone and tone-context contributions from the analysis pitch track, retaining the shared syllable contour and modeled utterance-level trends.

Cross-Fitted Adjustment. We divide episodes into five groups, fit the model on four groups, and adjust the remaining group. Each utterance is therefore adjusted using a model that has not seen its episode. We subtract the estimated component in semitone space, convert the adjusted track back to Hz, and repeat the same contour construction and scoring procedure used for raw pitch. The primary adjustment effect is the paired adjusted-minus-raw change in correlation on matched utterances.

Validation compares adjustment with real and shuffled tone labels and measures tone predictability before and after adjustment (Appendix A). We refer to the resulting tracks as tone-adjusted pitch.

3.6 Direct Tonal-Language Comparisons

We identify Chinese–Vietnamese, Chinese–Thai, and Vietnamese–Thai recordings through matching English recordings and English segment boundaries. Each recording retains its own timestamps. We apply SONAR filtering directly between the two dub transcripts and train direct bilingual alignments with paired TED2020 backgrounds.

We retain utterances with finite pitch scores for the direct tonal comparison and both corresponding English comparisons. Results are reported for the direct tonal pair, with clusters ordered by the first language. We adjust both recordings and compare raw and adjusted correlations on matched utterances, yielding 425 Chinese–Vietnamese, 279 Chinese–Thai, and 3,441 Vietnamese–Thai pairs.

3.7 Alignment Coverage and Cluster Ordering

We examine how coverage and aligned cluster ordering affect measured correspondence, using the French reference processed with the current settings. We define a cluster as a semantically aligned unit that may comprise one or more words in either language.

Coverage and Reordering. We measure coverage as the proportion of English words linked by FastAlign and the proportion of English speech time within analyzed clusters. English speech-time coverage provides a common measure across different target-language segmentations. We compare correlations within coverage bins and reweight their contributions to match the French coverage.

For utterances with at least two clusters, we measure reordering using normalized Kendall distance (Kendall, 1938) between the English and target cluster orders. A distance of zero indicates identical order and one complete reversal.

Semantic and Temporal Pairing. The main analysis uses semantic pairing: target clusters follow the order of their English counterparts. We compare this with temporal pairing, in which the same target clusters retain their own chronological order. The English sequence, selected clusters, acoustic frames, and resampling remain fixed across conditions.

We report mean correlations under both pairing conditions for utterances with at least two aligned clusters. We also calculate the paired temporal-minus-semantic correlation difference among reordered utterances, defined as those with nonzero reordering distance. When cluster orders agree, the two pairing conditions are identical. Detailed coverage and ordering analyses appear in Appendix B.

3.8 Temporal Feature Analysis

We measure utterance duration from timed transcript extent and derive phoneme counts using Phonemizer (Bernard and Titeux, 2021) with eSpeak NG (eSpeak NG contributors, 2024), and TLTK for Thai (Aroonmanakun, 2026). We retain pairs with valid outputs and equal positive sentence counts.

Duration and phoneme-count Spearman correlations are computed across utterances, separately from the within-utterance contour correlations. The phoneme counts represent canonical pronunciations derived from recognized text.

3.9 Contributions by Part of Speech

We assign alignment clusters to nouns, verbs, modifiers, or function words using their dominant English-side spaCy POS tags (Honnibal et al., 2020). For each group, its contribution is the full correlation minus the correlation after removing that group's frames from both contours; positive values indicate reduced correspondence. We repeat this analysis for raw and tone-adjusted pitch. Further methodological details, including the French reference sample and the non-additivity of contributions, are provided in Appendix C.

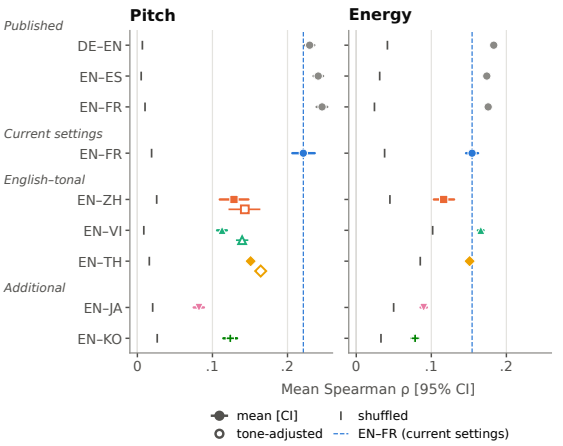


Figure 1: Mean pitch and energy correlations with 95% confidence intervals and shuffled baselines. Filled markers indicate raw correlations; open markers indicate tone-adjusted pitch. Vertical ticks mark baseline means.

4 Prosody Analysis Results

4.1 English–Tonal-Language Comparisons

We first compare English with the three languages that use lexical tone. Raw pitch correlations are lower for English–Mandarin Chinese (0.129), English–Vietnamese (0.113), and English–Thai (0.151) than for the English–French reference (0.221). The differences are substantial: even the highest English–tonal correlation is below the French value. We then estimate how much of this gap is associated with lexical tone by adjusting pitch for tone-related variation. The correlations increase to 0.143 for Chinese, 0.140 for Vietnamese, and 0.164 for Thai. These gains are positive but modest (+0.015, +0.027, and +0.013, respectively), and all three adjusted correlations remain below the French reference. Tone-related pitch variation therefore contributes to the difference, but adjusting for it does not close the gap. Lexical tone alone cannot account for the lower local pitch correspondence between English and these languages. Energy correlations show a different pattern: they are 0.116 for Chinese, 0.166 for Vietnamese, and 0.151 for Thai, compared with 0.154 for French. Thus, the lower correlations are not uniform across acoustic measures. The pitch results motivate examining lexical tone, while the energy results cau-

		Raw pitch		Pitch adjustment		Energy	
Pair	n	**Mean ρ [95% CI]**	**Baseline**	**Adj. ρ [95% CI]**	**Δ_{adjust} [95% CI]**	**Mean ρ [95% CI]**	**Baseline**
Published results: Xie et al. (2026)							
DE–EN	16,335	0.229 [0.223, 0.236]	0.007	—	—	0.183 [0.179, 0.187]	0.043
EN–ES	16,939	0.241 [0.235, 0.247]	0.005	—	—	0.174 [0.170, 0.178]	0.030
EN–FR	15,957	0.246 [0.239, 0.253]	0.010	—	—	0.176 [0.172, 0.179]	0.026
Current study: same-pipeline reference							
EN–FR	2,713	0.221 [0.206, 0.236]	0.019	—	—	0.154 [0.146, 0.162]	0.038
Current study: English–tonal pairs							
EN–ZH	1,838	0.129 [0.110, 0.148]	0.026	0.143 [0.123, 0.163]	+0.015 [0.003, 0.025]	0.116 [0.103, 0.130]	0.045
EN–VI	22,332	0.113 [0.106, 0.119]	0.009	0.140 [0.132, 0.147]	+0.027 [0.023, 0.030]	0.166 [0.162, 0.170]	0.102
EN–TH	31,261	0.151 [0.146, 0.155]	0.016	0.164 [0.160, 0.169]	+0.013 [0.012, 0.015]	0.151 [0.148, 0.153]	0.085
Current study: Japanese and Korean comparisons							
EN–JA	15,267	0.082 [0.075, 0.089]	0.020	—	—	0.090 [0.086, 0.094]	0.050
EN–KO	11,350	0.124 [0.115, 0.133]	0.026	—	—	0.078 [0.073, 0.083]	0.033
Current study: tonal–tonal pairs							
ZH–VI	425	0.063 [0.023, 0.103]	0.001	0.096 [0.058, 0.130]	+0.033 [0.006, 0.056]	0.130 [0.113, 0.151]	0.075
ZH–TH	279	0.149 [0.106, 0.190]	0.006	0.149 [0.104, 0.193]	+0.000 [−0.026, 0.023]	0.092 [0.072, 0.111]	0.052
VI–TH	3,441	0.088 [0.076, 0.100]	0.007	0.150 [0.137, 0.163]	+0.062 [0.052, 0.072]	0.190 [0.182, 0.197]	0.136

Table 4: Mean utterance-level correlations with 95% confidence intervals and unit-shuffled baselines. Current-study intervals use episode-clustered bootstrapping. The current French reference uses the same processing settings as the new language pairs. Δ_{adjust} is the paired adjusted-minus-raw pitch change. For English–tonal pairs, n denotes the raw cohort.

tion against treating the difference as a general reduction in all prosodic correspondence.

4.2 Tonal–Tonal Comparisons

The three tonal-language pairs show that sharing a lexical tone structure does not by itself yield strong local pitch correspondence. Raw correlations range from 0.063 for Chinese–Vietnamese to 0.149 for Chinese–Thai, with Vietnamese–Thai at 0.088. Tone adjustment increases the Chinese–Vietnamese correlation to 0.096 and the Vietnamese–Thai correlation to 0.150, but leaves the Chinese–Thai correlation essentially unchanged at 0.149. The effect of adjustment therefore varies across pairs, and the resulting correlations remain modest.

These results suggest that lexical tone contributes to cross-lingual pitch differences, but does not account for them on its own. Even when both languages use lexical tone, their local pitch patterns do not necessarily correspond. Other differences between the languages and between the structures of translated utterances must also be considered.

4.3 English–Japanese and English–Korean Comparisons

The English–Japanese and English–Korean pairs provide a further comparison with languages that do not use lexical tone. Japanese uses lexical pitch accent, while Korean does not use lexical tone. Their raw pitch correlations with English are 0.082 and 0.124, respectively; their energy correlations are 0.090 and 0.078. The low pitch correlations for these pairs show that lexical tone is not necessary for weak cross-lingual local pitch correspondence. Taken together with the tonal-language results, this comparison indicates that tone status alone does not explain the observed differences.

These pairs also differ more strongly from English in the ordering of semantically aligned words and have lower alignment coverage. This co-occurrence motivates examining syntactic word-order divergence as an indicator of broader differences in sentence structure.

4.4 Word Reordering and Alignment Coverage

The mean normalized reordering distance is greatest for Japanese (0.241) and Korean (0.151), compared with French (0.045) and the tonal-language pairs (0.063–0.094). Alignment coverage follows a similar broad pattern: English speech-time coverage is 52% for Japanese and 46% for Korean, compared with 62–66% for the tonal-language pairs and 72% for French. Word coverage is also lower for Japanese and Korean (42% and 39%) than for French (67%) or the tonal-language pairs (55–60%). These measures show that translated utterances differ not only in the ordering of aligned content but also in how much content can be aligned.

We compare semantic pairing, which brings aligned words together, with chronological pairing, which preserves their original order in each recording. The displayed semantic-pairing pitch correlations are 0.221 for French, 0.132 for Chinese, 0.113 for Vietnamese, 0.151 for Thai, 0.083 for Japanese, and 0.130 for Korean. For Japanese and

Pair	Mean R	Semantic ρ	Temporal ρ	Temporal advantage [95% CI]
EN–FR	0.045	0.221	0.227	+0.013 [−0.001, 0.025]
EN–ZH	0.094	0.132	0.144	+0.025 [0.001, 0.049]
EN–VI	0.063	0.113	0.121	+0.015 [0.010, 0.021]
EN–TH	0.072	0.151	0.169	+0.033 [0.029, 0.037]
EN–JA	0.241	0.083	0.141	+0.090 [0.080, 0.101]
EN–KO	0.151	0.130	0.183	+0.116 [0.103, 0.129]

Table 5: Cluster reordering and raw-pitch correspondence under semantic and temporal pairing. R is normalized Kendall distance between cluster orders, ranging from zero for identical order to one for complete reversal. Mean R and mean correlations use utterances with at least two aligned clusters; this restriction accounts for differences from Table 4. Temporal advantage is the mean paired $\rho_{\text{temporal}} - \rho_{\text{semantic}}$ among reordered utterances ($R > 0$), rather than the difference between the displayed corpus means. Intervals use episode-clustered bootstrapping.

Korean, the corresponding chronological-pairing correlations are higher: 0.141 and 0.183. Among reordered utterances, the mean paired changes from semantic to chronological pairing are +0.090 for Japanese and +0.116 for Korean. The changes are smaller for French (+0.013), Chinese (+0.025), Vietnamese (+0.015), and Thai (+0.033).

For Japanese and Korean, then, semantic reordering substantially lowers the measured correlation relative to preserving the original sentence order. Aligning words by meaning does not recover strong local prosodic correspondence in these pairs. Instead, it produces much weaker correlations, consistent with the possibility that semantically corresponding words occur in different prosodic contexts when sentence structures differ. The higher correlations under chronological pairing suggest that some broader sentence-level pitch organization remains shared; they do not show that local prosody corresponds at the semantic word level.

4.5 Summary

Tone adjustment yields modest pitch-correspondence gains for English–tonal pairs and larger gains for two tonal–tonal pairs, but correlations remain modest. Low correspondence also occurs for English–Japanese and English–Korean, indicating that the pattern is not specific to lexical tone. Japanese and Korean show greater word-order divergence and lower alignment coverage, and their correlations are substantially higher under temporal than semantic pairing. Their greater word-order divergence indicates greater syntactic differences, while the low semantic-pairing correlations suggest that fine-grained prosodic correspondence may be weaker across structurally different language pairs. The higher temporal-pairing correlations may reflect shared utterance-level pitch structure rather than correspondence between semantically aligned words. Appendices C and D.3 provide complementary analyses of POS contributions and utterance duration correspondence.

5 Conclusion

We analyzed word-level prosodic correspondence in professionally dubbed speech across five English–target language pairs and three pairs among tonal languages. All pairs show positive mean pitch and energy correlations above unit-shuffled baselines. Pitch correspondence for the English–target pairs remains below European-language estimates, while energy shows a more mixed pattern. Tone adjustment improves all three English–tonal pitch correlations but does not close the gap; low correlations for English–Japanese and English–Korean also show that weak correspondence is not specific to lexical tone.

Japanese and Korean show the greatest word-order divergence and lowest alignment coverage. Their correlations are substantially higher under temporal than semantic pairing, indicating that semantic alignment does not recover local prosodic correspondence in these pairs. The greater word-order divergence reflects broader syntactic differences, which are associated with weaker fine-grained prosodic correspondence. The higher correlations under temporal pairing may instead reflect shared utterance-level pitch structure.

Together, our findings indicate that lexical tone contributes to cross-lingual pitch differences, but does not account for them on its own. Local prosodic correspondence may be weak across languages with substantially different syntactic structures, even when some broader utterance-level correspondence remains. This motivates expressive speech translation approaches that consider both sentence structure and their prosodic context.

Limitations

Licensing restrictions prevent us from sharing the speech data, but our detailed pipeline allows researchers to repeat the analyses on comparable dubbed speech. Acting, translation choices, and timing constraints may limit generalization beyond television dubbing. Samples differ in size, content, speakers, and production practices, especially for direct tonal-language comparisons. Using the same pipeline for French does not remove these differences, so the observed gaps cannot be explained by language structure alone.

Automatic processing may introduce errors that differ across languages. Our gold alignment evaluations cover only Chinese and Japanese text outside the dubbing domain, so accuracy on the recordings remains uncertain. Coverage analyses and sensitivity checks cannot rule out systematic errors or effects of filtering.

Tone adjustment does not completely separate lexical tone from intonation. Detrending also removes meaningful pitch patterns tied to sentence position. The reordering analyses suggest relationships rather than establish causes. Higher correlations under temporal pairing may reflect shared sentence-level patterns without showing that corresponding words carry similar prosody.

Listening studies are needed to determine whether emphasis, emotion, and communicative intent are preserved. Episode-clustered confidence intervals may not fully account for shared casts, production practices, or English recordings across language pairs.

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A Lexical Tone Adjustment Validation

A.1 Tone-Label-Shuffled Baseline

The main results report paired changes from raw to tone-adjusted pitch. Here, we test whether the gains depend on the tone labels by fitting the same adjustment model with shuffled labels, using the same episode-based cross-fitting procedure. We denote the paired correlation difference between real-label and shuffled-label adjustment by Δ_{label}. This baseline evaluates the adjustment procedure; the unit-shuffled baseline instead permutes target alignment clusters before correlation is measured.

Real-label adjustment exceeds the tone-label-shuffled baseline by 0.016 for Chinese, 0.027 for Vietnamese, and 0.014 for Thai, with all three confidence intervals excluding zero (Table 6). The gains therefore depend on information carried by the tone labels. This comparison supports the specificity of the adjustment, without requiring a complete separation of tone and intonation.

Pair	Δ_{label} **[95% CI]**
EN–ZH	+0.016 [0.005, 0.026]
EN–VI	+0.027 [0.024, 0.030]
EN–TH	+0.014 [0.012, 0.016]

Table 6: Paired differences between real-label and shuffled-label tone adjustment on matched eligible utterances. Intervals use episode-clustered bootstrapping. These contrasts differ from the adjusted-minus-raw changes in the main results.

A.2 Tonal-Pair Label Baseline

For the direct tonal-language pairs, we compare adjustment of both recordings using real labels with adjustment of both recordings using shuffled labels. The point differences are 0.035 for Chinese–Vietnamese, 0.005 for Chinese–Thai, and 0.063 for Vietnamese–Thai (Table 7). Their ordering agrees with the adjusted-minus-raw changes reported in Table 4.

Pair	**Matched** n	Δ_{label}
ZH–VI	425	+0.035
ZH–TH	279	+0.005
VI–TH	3,441	+0.063

Table 7: Real-label-minus-shuffled-label pitch correlation differences, with both recordings adjusted on matched utterances. Only point estimates are reported for these contrasts; the adjusted-minus-raw intervals in the main results refer to a different comparison.

A.3 Tone Contours and Predictability

Figure 2 compares tone-conditioned syllable contours before and after adjustment. Raw contours show differences in pitch level and shape across tone categories. These differences are reduced after adjustment, although some separation between categories remains.

We also measure tone predictability from syllable contours using the 3,000-utterance diagnostic sample. Balanced accuracy decreases from 0.427 to 0.301 for Chinese, from 0.444 to 0.287 for Vietnamese, and from 0.464 to 0.333 for Thai (Table 8). These changes correspond to reductions of approximately 55%, 57%, and 50% in above-chance balanced accuracy. Adjusted accuracy remains above chance in all three languages. The adjustment therefore reduces tone predictability without eliminating it; the percentages describe reductions in classification performance, rather than fractions of tone information removed.

Language	**Raw**	**Adjusted**	**Chance**
Chinese	0.427	0.301	0.200
Vietnamese	0.444	0.287	0.167
Thai	0.464	0.333	0.200

Table 8: Tone-classification balanced accuracy from raw and adjusted syllable contours on the diagnostic sample.

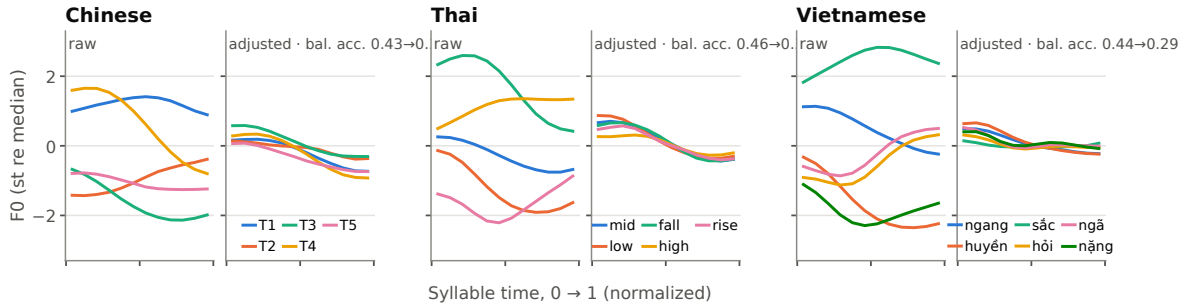


Figure 2: Tone-conditioned syllable pitch contours before and after adjustment for Chinese, Thai, and Vietnamese. Pitch is expressed in semitones relative to the utterance median, with syllable time normalized from zero to one. Colors identify tone categories within each language. The annotations report tone-classification balanced accuracy before and after adjustment.

B Alignment and Reordering Sensitivity

B.1 Alignment Coverage

Table 9 reports coverage and cluster ordering for the raw-pitch runs. French uses the current processing settings throughout this section. FastAlign links 67% of English words for French, compared with 55–60% for the three tonal languages, 42% for Japanese, and 39% for Korean. English speech-time coverage follows the same pattern. Japanese and Korean also have fewer clusters per utterance, averaging 4.1 and 3.4, respectively, compared with 8.0 for French.

Pitch correlation generally increases with coverage. For Japanese, the mean rises from 0.070 in the lowest coverage bin to 0.115 in the highest. At 60–80% English speech-time coverage, French reaches 0.223, compared with 0.094–0.155 for the Asian-language pairs. Reweighting each pair to the French coverage distribution increases the Japanese and Korean means by approximately 0.015 and the tonal means by less than 0.005. These point estimates show that matching the coverage distribution does not eliminate the difference.

B.2 Cluster Reordering

We measure reordering by comparing the English order of aligned clusters with their order in the target recording. The frequency of any reordering and its extent capture different properties: French and Korean have similar proportions of reordered utterances, at 48% and 46%, but mean distances of 0.045 and 0.151, respectively. Japanese has the largest mean distance, at 0.241, compared with 0.063–0.094 for the English–tonal pairs.

These statistics describe automatically recovered alignments. Missed links can suppress observed crossings, while incorrect links can introduce them. Utterances with fewer than two clusters are excluded from the reordering analysis; 12% of Korean utterances contain only one cluster.

B.3 Semantic and Temporal Pairing

We compare two pairings of the same aligned clusters. Semantic pairing places target clusters in the order of their English counterparts, as in the main analysis. Temporal pairing retains the target clusters' chronological order while leaving the English sequence fixed. The selected clusters, acoustic frames, and resampling procedure are identical, so this comparison changes ordering while holding coverage constant.

We define the temporal advantage as $D = \rho_{\text{temporal}} - \rho_{\text{semantic}}$. Table 5 in the main results reports raw correlations and paired differences. Table 10 compares the paired differences before and after detrending among reordered utterances ($R > 0$) with at least two clusters. The pairing conditions coincide when the cluster orders agree.

Japanese and Korean have the largest paired temporal advantages, at 0.090 and 0.116, respectively. The gains are 0.025 for Chinese, 0.015 for Vietnamese, 0.033 for Thai, and 0.013 for French. All intervals except the French interval exclude zero.

B.4 Within-Utterance Pitch Trends

Table 11 provides the full pitch-trend statistics summarized in Table 13. Each recording's trend is fitted in its own temporal cluster order, using utterances with at least three clusters. The per-cluster slope depends on the number of clusters, whereas the fitted first-to-last change summarizes the trend across the analyzed utterance.

Pair	English coverage		Clusters/utt.	Reordering	
	Words	Speech time		Any	Mean R
EN–FR	67%	72%	8.0	48%	0.045
EN–ZH	55%	62%	5.2	48%	0.094
EN–VI	60%	66%	6.4	50%	0.063
EN–TH	59%	66%	6.4	53%	0.072
EN–JA	42%	52%	4.1	64%	0.241
EN–KO	39%	46%	3.4	46%	0.151

Table 9: Alignment coverage and cluster reordering in the raw-pitch runs. Word coverage is the proportion of English words linked by FastAlign; speech-time coverage is the proportion of English speech time within analyzed clusters. Reordering statistics use utterances with at least two clusters. "Any" denotes the proportion with $R > 0$, where R is normalized Kendall distance between cluster orders. Zero indicates identical order and one complete reversal.

Pair	Temporal advantage D [95% CI]	
	Before detrending	After detrending
EN–FR	+0.013 [−0.001, 0.025]	+0.000 [−0.015, 0.015]
EN–ZH	+0.025 [0.001, 0.049]	+0.007 [−0.017, 0.032]
EN–VI	+0.015 [0.010, 0.021]	+0.007 [0.001, 0.013]
EN–TH	+0.033 [0.029, 0.037]	+0.018 [0.013, 0.023]
EN–JA	+0.090 [0.080, 0.101]	+0.005 [−0.004, 0.014]
EN–KO	+0.116 [0.103, 0.129]	+0.023 [0.010, 0.035]

Table 10: Paired temporal-minus-semantic pitch correlation differences among reordered utterances ($R > 0$) with at least two aligned clusters, before and after removing each recording's linear pitch trend. Both conditions use the same selected clusters. Intervals use episode-clustered bootstrapping. French uses the current processing settings.

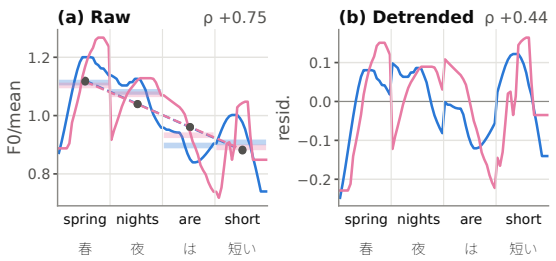


Figure 3: Illustration of broad pitch trends and residual correspondence. (a) Raw English and Japanese contours under semantic pairing, with fitted temporal trends shown by dashed lines. (b) Paired contours after removing each recording's temporal trend. Blue denotes English and pink Japanese.

Mean first-to-last changes are negative on both sides of every language pair, with all corresponding confidence intervals excluding zero. Between 65% and 76% of utterances have falling raw-pitch trends. Declining trends also remain after tone adjustment, with mean first-to-last changes of -10.7% for Chinese, -6.8% for Vietnamese, and -14.0% for Thai. Tone adjustment therefore retains broad sentence-level pitch decline.

B.5 Detrending Sensitivity

Figure 4 shows how the temporal advantage varies with reordering distance before and after detrending. On each side, we remove a linear trend fitted to cluster means over the clusters' positions in that recording's own temporal order. We then repeat both pairing conditions.

Table 13 reports semantic-pairing correlations before and after trend removal, together with the raw first-to-last pitch changes. Mean correlations decrease for every pair: French falls from 0.222 to 0.123, while the Asian-language pairs fall from 0.084–0.153 to 0.046–0.062. Japanese and Korean retain means of 0.051 and 0.060, respectively, within the tonal-language range; French remains higher. Figure 3 illustrates a shared declining trend and positive residual correspondence in an English–Japanese example.

The temporal advantage is substantially smaller after detrending, particularly for Japanese and Korean. Their mean paired advantages decrease from 0.090 to 0.005 and from 0.116 to 0.023, respectively (Table 10). The Japanese interval includes zero after detrending, while positive advantages with intervals excluding zero remain for Vietnamese, Thai, and Korean. Removing broad linear trends therefore reduces sensitivity to cluster ordering, although the pairing conditions do not become identical in every language.

The results support the interpretation that semantic realignment can disrupt shared sentence-level pitch trends. These trends are meaningful prosodic structure, and detrending also removes genuine position-linked effects such as final lower-

Pair	n	Mean clusters	Recording	Slope per cluster (% of mean F_0)	Fitted first-to-last change		Falling (%)	R^2
					% of mean F_0	Semitones		
EN–FR	2,595	8.3	English	−2.9 [−3.1, −2.6]	−14.5 [−15.7, −13.4]	−2.58 [−2.78, −2.39]	76	0.42
			French	−2.5 [−2.8, −2.2]	−11.8 [−13.0, −10.6]	−2.13 [−2.33, −1.92]	72	0.41
EN–ZH	1,627	5.6	English	−3.5 [−4.0, −3.0]	−12.3 [−13.7, −10.7]	−2.16 [−2.43, −1.89]	70	0.48
			Chinese	−3.0 [−3.4, −2.6]	−10.1 [−11.3, −8.9]	−1.81 [−2.03, −1.59]	68	0.41
			Chinese (adjusted)	−3.3 [−3.6, −2.9]	−10.7 [−11.9, −9.5]	−1.92 [−2.13, −1.71]	70	0.45
EN–VI	20,590	6.7	English	−3.3 [−3.4, −3.2]	−14.1 [−14.5, −13.6]	−2.50 [−2.57, −2.42]	76	0.46
			Vietnamese	−1.7 [−1.8, −1.5]	−6.8 [−7.3, −6.4]	−1.23 [−1.31, −1.15]	65	0.32
			Vietnamese (adjusted)	−1.6 [−1.8, −1.5]	−6.8 [−7.2, −6.4]	−1.21 [−1.29, −1.13]	69	0.38
EN–TH	28,817	6.8	English	−3.3 [−3.4, −3.2]	−13.8 [−14.2, −13.5]	−2.44 [−2.51, −2.39]	76	0.46
			Thai	−3.7 [−3.9, −3.6]	−14.8 [−15.1, −14.4]	−2.58 [−2.65, −2.52]	74	0.38
			Thai (adjusted)	−3.5 [−3.6, −3.4]	−14.0 [−14.3, −13.6]	−2.45 [−2.51, −2.39]	75	0.42
EN–JA	11,217	4.9	English	−4.1 [−4.3, −3.9]	−12.7 [−13.3, −12.2]	−2.24 [−2.35, −2.14]	74	0.51
			Japanese	−4.3 [−4.5, −4.1]	−12.6 [−13.1, −12.0]	−2.25 [−2.35, −2.15]	69	0.46
EN–KO	6,875	4.4	English	−4.3 [−4.5, −4.1]	−12.1 [−12.7, −11.5]	−2.13 [−2.23, −2.02]	75	0.51
			Korean	−4.2 [−4.5, −3.9]	−11.9 [−12.6, −11.2]	−2.08 [−2.21, −1.95]	73	0.51

Table 11: Pitch trends estimated from utterances with at least three aligned clusters. For each recording, we fit a straight line through the cluster mean pitches in their original temporal order. The per-cluster slope measures the fitted pitch change between successive clusters, expressed as a percentage of the recording's utterance-mean F_0. Because cluster counts differ across languages, these slopes should not be compared directly. The first-to-last columns instead summarize the fitted change across the analyzed utterance, from its first to its last cluster, providing an utterance-wide measure for comparison. This change is reported both as a percentage of mean F_0 and in semitones, a logarithmic pitch scale. Negative values indicate falling pitch. "Falling" gives the percentage of utterances with a negative fitted slope, while R^2 describes the proportion of variation in cluster mean pitch explained by the line. Brackets give episode-clustered 95% confidence intervals for the mean slopes and changes. Adjusted rows use tone-adjusted pitch; French uses the current processing settings.

ing. It is therefore a sensitivity analysis, rather than a complete separation of sentence-level and local prosody. Residual semantic-pairing means use utterances with at least three clusters: with two clusters, the fitted trend removes all between-cluster mean differences. The detrended Asian-language means lie within a similar range, while the French mean remains higher.

C Part-of-Speech Contributions

We examine how word classes contribute to word-aligned pitch correspondence. Alignment clusters are assigned to nouns, verbs, modifiers, or function words using their dominant English-side POS tags from spaCy. For each group, we remove its frames from both contours and subtract the resulting correlation from the full correlation. A positive contribution indicates that removing the group reduces correspondence. Groups are evaluated separately, so their contributions are not additive. We repeat the analysis before and after lexical-tone adjustment. The French comparison uses a separate 10,000-utterance reference sample processed with the same settings.

Figure 5 summarizes the results, and Table 12 reports the corresponding values. Before adjustment, verbs contribute most in Chinese, function words in Vietnamese, and nouns in Thai. After adjustment, noun contributions increase from 0.003 to 0.013 in Chinese, from 0.010 to 0.017 in Vietnamese, and from 0.016 to 0.018 in Thai. All three adjusted noun contributions have 95% confidence intervals excluding zero and are numerically close to or above the French value of 0.015.

Modifier contributions also increase in all three tonal languages. Function-word contributions decrease most in Chinese, from 0.016 to 0.007, with the adjusted interval including zero. Verb contributions decrease in Chinese and increase in Vietnamese and Thai. These changes describe how tone adjustment shifts the distribution of POS contributions; the reported intervals assess individual contributions, not the significance of raw-to-adjusted changes.

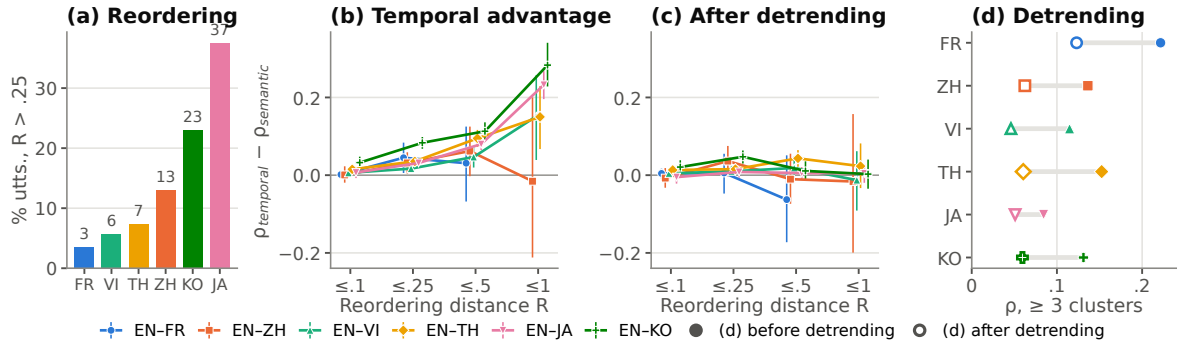


Figure 4: Cluster reordering and detrending sensitivity. (a) Proportion of utterances with normalized reordering distance $R > 0.25$. (b) Temporal-minus-semantic pitch correlation differences across reordering-distance bins. (c) The same comparison after removing linear trends fitted to cluster means in each recording's temporal order. (d) Mean semantic-pairing correlations before and after detrending. Reordering comparisons use utterances with at least two clusters; panel d uses utterances with at least three. French uses the current processing settings.

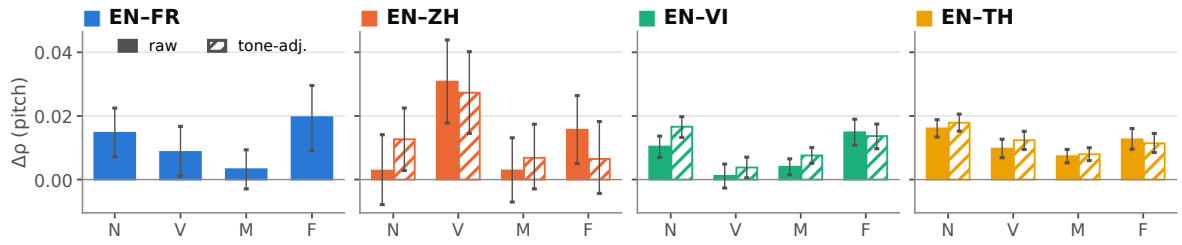


Figure 5: POS contributions to pitch correspondence before and after tone adjustment. Each contribution is the full correlation minus the correlation after removing the group's frames from both contours. N, V, M, and F denote nouns, verbs, modifiers, and function words, respectively. Error bars indicate 95% confidence intervals for individual contributions. French uses the current processing settings.

Pair / condition	Nouns	Verbs	Mod.	Func.
EN–FR	+.015*	+.009*	+.003	+.020*
EN–ZH	+.003	+.031*	+.003	+.016*
Adjusted	+.013*	+.027*	+.007	+.007
EN–VI	+.010*	+.001	+.004*	+.015*
Adjusted	+.017*	+.004*	+.008*	+.014*
EN–TH	+.016*	+.010*	+.007*	+.013*
Adjusted	+.018*	+.012*	+.008*	+.011*

Table 12: Leave-one-group-out pitch contributions. Unmarked rows use raw pitch; adjusted rows use tone-adjusted pitch. An asterisk indicates that the individual contribution's 95% confidence interval excludes zero. It does not test the change between raw and adjusted contributions. Mod. and Func. denote modifiers and function words. French uses the current processing settings on the separate POS reference sample.

D Additional Correlation Analyses

D.1 Utterance-Level Distributions

Figure 6 shows the distributions of raw pitch and energy correlations across utterances. The distributions contain both positive and negative scores, despite their positive means. The vertical lines distinguish each pair's mean from its unit-shuffled baseline. The annotations report the mean and the mean difference above baseline; the latter is distinct from the tone-adjustment change used in the main results.

D.2 Matched Tonal-Pair Comparisons

Figure 7 shows the direct tonal-pair correlations alongside the corresponding English comparisons on the matched cohorts. English comparisons provide context for the direct-pair analysis; each language pair is aligned separately. The correlations therefore describe pair-specific contour constructions, rather than a single alignment shared by all three recordings.

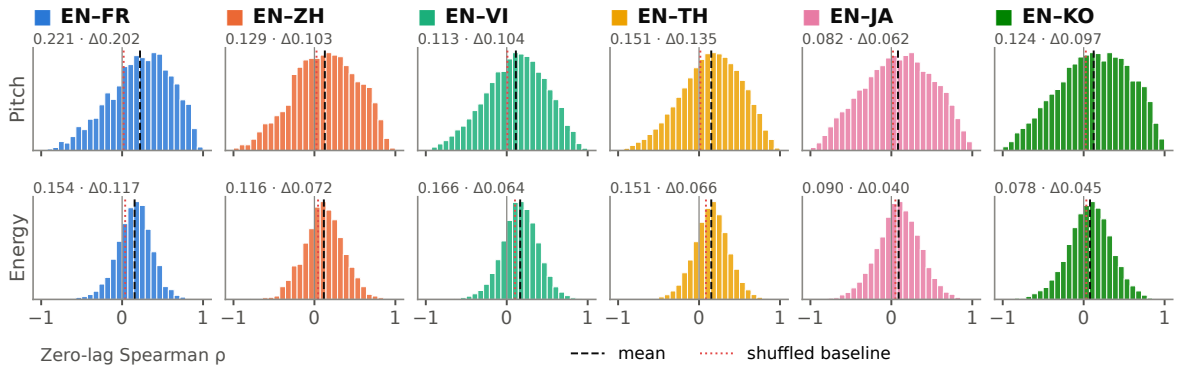


Figure 6: Distributions of utterance-level raw pitch and energy correlations. Black dashed lines mark mean correlations, and red dotted lines mark unit-shuffled baseline means. Each annotation gives the mean correlation and its difference above baseline. These differences use unrounded estimates. French uses the current processing settings.

	First-to-last pitch change (%)		Pitch correlation (mean ρ)	
Pair	**EN**	**Other**	**Before**	**After**
EN–FR	−14.5	−11.8	0.222	0.123
EN–ZH	−12.3	−10.1	0.136	0.062
EN–VI	−14.1	−6.8	0.115	0.046
EN–TH	−13.8	−14.8	0.153	0.060
EN–JA	−12.7	−12.6	0.084	0.051
EN–KO	−12.1	−11.9	0.131	0.060

Table 13: Within-utterance pitch decline and correspondence before and after detrending. Pitch change is the mean fitted first-to-last change as a percentage of utterance-mean F_0, reported separately for English (EN) and the other language; negative values indicate decline. Each recording's trend is fitted through its cluster mean pitches in its own temporal order. The final columns report mean semantic-pairing correlations before and after removing these trends. Each pair uses its own utterances with at least three aligned clusters, with the same utterances scored before and after detrending.

	Duration		**Phone count**	
Pair	n	ρ	n	ρ
EN–FR	1,931	0.938	1,931	0.905
EN–ZH	1,318	0.758	1,318	0.730
EN–VI	17,452	0.916	17,452	0.888
EN–TH	21,183	0.847	21,120	0.837

Table 14: Cross-utterance Spearman correlations in duration and canonical phoneme count at the primary SONAR threshold of 0.45. These are correlations across utterances, rather than means of within-utterance contour correlations. French uses the current processing settings.

D.3 Duration and Phoneme Counts

Table 14 reports cross-utterance correlations in duration and canonical phoneme count. Duration is measured from timed transcript extent, and phoneme counts come from grapheme-to-phoneme conversion of the recognized text. We retain pairs with equal positive sentence counts for these comparisons. The Thai phoneme-count cohort is smaller because some pairs lack valid counts.

Duration correlations range from 0.758 to 0.916 for the English–tonal pairs, and phoneme-count correlations from 0.730 to 0.888. These results indicate correspondence in coarse temporal and transcript-derived quantities alongside the lower contour correlations. Canonical counts describe expected pronunciations of the recognized text, rather than acoustically measured realized phonemes.

E Filtering and Feature Sensitivity

E.1 Semantic Filtering Sensitivity

We repeat the pitch analysis at SONAR 0.63 rather than the primary threshold of 0.45. This is the semantic threshold used by Xie et al. (2026). Raw and adjusted means for the tonal languages remain below the current French reference at this threshold (Table 15). The descriptive difference therefore persists under stricter semantic filtering.

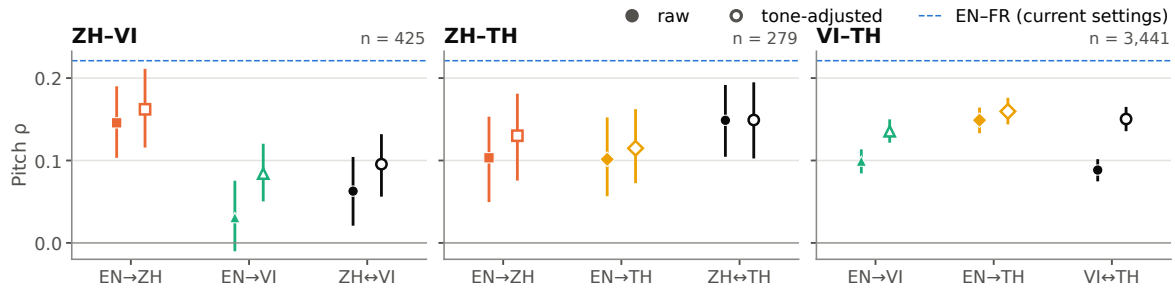


Figure 7: Raw and tone-adjusted pitch correlations for matched tonal-pair cohorts and their corresponding English comparisons. Filled markers indicate raw pitch; open markers indicate adjusted pitch. Both recordings are adjusted in the direct tonal-pair comparisons, while only the tonal recording is adjusted in the English comparisons. Each language pair is aligned separately. Dashed lines mark the current French reference, which is outside these cohorts.

Pair	Raw ρ	Adjusted ρ
EN–FR	0.225	—
EN–ZH	0.154	0.163
EN–VI	0.123	0.153
EN–TH	0.158	0.174

Table 15: Mean pitch correlations at SONAR 0.63. Raw and adjusted values use their respective available scoring cohorts; their differences are not paired adjustment estimates. French uses the current processing settings.

E.2 Alignment Benchmarks

Table 16 reports precision, recall, and alignment error rate against human gold links. The Japanese benchmark (Neubig, 2011) has lower recall than the Chinese benchmark (Liu and Sun, 2015), making missed links a particular concern for the Japanese contour analysis.

Language	n	Precision	Recall	AER
Chinese	900	80.47%	48.52%	39.42%
Japanese	1,235	70.95%	22.23%	66.15%

Table 16: Alignment performance against human gold links in text benchmarks. AER denotes alignment error rate. n is the number of benchmark sentences. The Chinese benchmark contains newswire and the Japanese benchmark Wikipedia text.

These evaluations use text outside the dubbing domain. Benchmark sentence text is included in unsupervised alignment training, while gold links are withheld. The scores characterize this transductive benchmark setting and do not directly measure alignment accuracy in the analyzed recordings.

E.3 Contour Resolution and Track Checks

Varying contour resolution from 10 to 40 frames per aligned cluster changes raw-pitch mean correlations by at most approximately 0.0021 in the reported checks. The Thai and Vietnamese checks use samples of roughly 3,000 pairs. Adjusted-pitch resolution checks show a spread of at most 0.002.

A scan of 169,522 saved raw-pitch tracks identifies five containing negative interpolated pitch values. Excluding the affected utterance pairs changes the corresponding mean correlations by less than 0.0002. Checks on 300 utterances per language also confirm that tone adjustment preserves time coordinates and voicing, and leaves frames outside labeled syllables unchanged.

E.4 Contour Construction Example

Figure 8 illustrates contour construction for a Chinese–English utterance. It shows the syllable intervals used for tone adjustment, the timed English transcript, and the selected alignment clusters in English order. The example illustrates the automatic procedure rather than a manually corrected alignment.

E.5 Statistical Dependence

Confidence intervals use episode-clustered bootstrapping, retaining the utterances within each sampled episode together. Episodes can also share casts and recording conditions, and English segments can recur across language comparisons. These dependencies beyond the episode are not fully represented by the reported intervals or utterance-level rank tests.

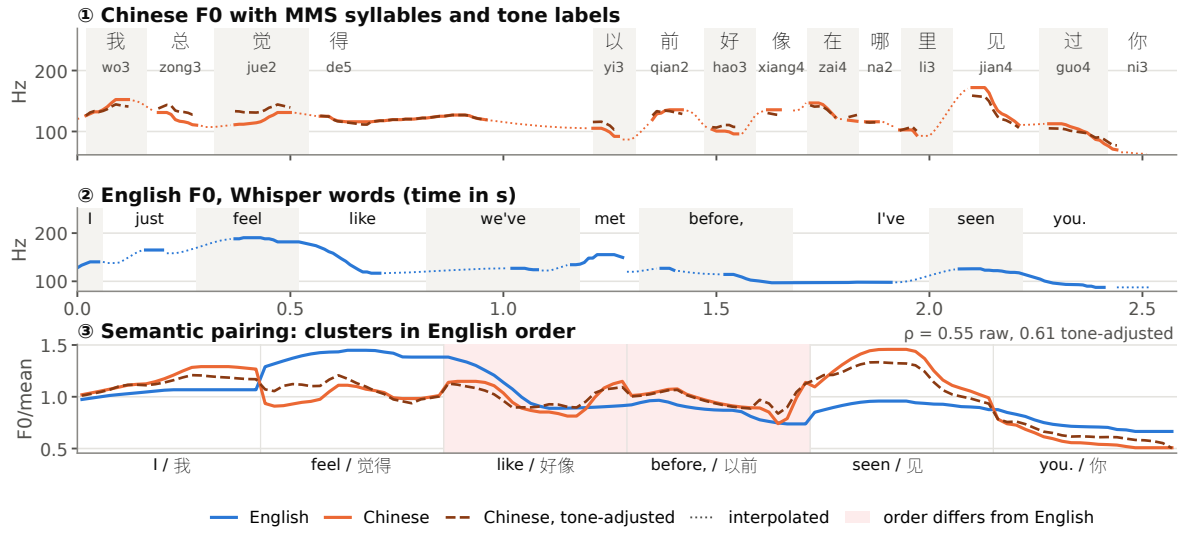


Figure 8: Chinese–English contour construction. Top: Chinese pitch with MMS syllable intervals and tone labels. Middle: English pitch with timed transcript units. Bottom: normalized contours for selected alignment clusters in English order, including raw and adjusted Chinese pitch. Shading marks clusters whose order differs between recordings; dotted segments indicate interpolated pitch. Displayed links are automatically generated.

F Estimated Voice-Label Distribution

We use inaSpeechSegmenter (Doukhan et al., 2018) obtain automatic male/female voice labels for both recordings in each scored English–tonal pair and the English–French reference. Table 17 summarizes the distributions. These are acoustic classifications, not self-reported gender identities. Processing completed without errors; percentages exclude recordings with no detected speech (4–96 per language side). Japanese and Korean have not yet been evaluated.

Pair	English M/F	Target M/F	Pairs
EN–FR	55.1 / 44.9	54.0 / 46.0	2,713
EN–ZH	62.7 / 37.3	61.9 / 38.1	1,838
EN–VI	57.2 / 42.8	56.2 / 43.8	22,332
EN–TH	57.0 / 43.0	55.6 / 44.4	31,261

Table 17: Automatic male/female (M/F) voice-label percentages for the scored utterance pairs. Percentages exclude recordings with no detected speech; pair counts include the full scored cohorts. French uses the current processing settings.

English and target labels agree for approximately 94–95% of pairs. Across the evaluated language sides, median F_0 ranges from 113–137 Hz for male-labeled recordings and 180–225 Hz for female-labeled recordings. These summaries provide descriptive checks on the automatic labels; they do not establish classification accuracy or control for speaker composition.